EX05 - Sum of Squares - for, range and Accumulators

Beginner Step-by-Step Guide

This guide follows the real file `ex05_sum_squares.py`. It explains the problem first and then the code from the perspective of someone starting Python.

1. What we want to solve

Compute 1^2+2^2+...+N^2 with a loop and compare it with the closed-form formula.

Problem variables / elements

Name	Meaning
N	last integer included
sum_squares	running sum

Equation or logic on paper

```
S = sum(i**2), i=1..N
S = N*(N+1)*(2*N+1)/6
```

Manual check

For N=3: 1+4+9=14. The formula gives 3*4*7/6=14.

2. What we already know and what is new

Already seen:

- variables
- if
- isinstance()

New in this exercise:

- for
- range(start,stop)
- accumulator
- +=
- //
- iterative vs analytical validation

3. New Python concepts

1. for loop

Syntax: `for i in range(1, N + 1):`

for takes one value from the iterable on each iteration. Here i runs through 1,2,...,N.

2. range

Syntax: `range(1, N + 1)`

range includes start and excludes stop; N+1 is used so N is included.

3. Accumulator

Syntax: `sum_squares = 0`

An accumulator stores the partial result. A sum starts at 0 because 0 is the additive identity.

4. Augmented assignment

Syntax: `sum_squares += i**2`

Equivalent to sum_squares = sum_squares + i**2.

5. Integer division

Syntax: `//`

// produces an integer quotient. In this formula the final result is an integer.

4. Code step by step

Step 1. Define N and accumulator

```
N = 1000000
sum_squares = 0
```

N determines how many terms are summed. The accumulator starts at 0.

Step 2. Validate N

```
if not isinstance(N, int) or N < 1:
    ...
```

not flips the Boolean and or means either invalid condition rejects N.

Step 3. Loop through integers

```
for i in range(1, N + 1):
    sum_squares += i**2
```

Each iteration computes i**2 and adds it to the previous total.

Step 4. Compute reference

```
analytical_result = (N * (N + 1) * (2 * N + 1)) // 6
```

The closed form uses a different path from the loop, so it acts as validation.

5. Execution flow

```
N
 -> validate
 -> sum_squares=0
 -> for i=1..N
 -> add i**2
 -> compare analytical
```

6. Expected result / check

With N=1,000,000 both source results are 333333833333500000.

7. Common beginner mistakes

- Using range(1,N) and excluding N.
- Starting the accumulator with the wrong value.
- Confusing += with =+.
- Not checking that N is int.

8. If you remember only five things

- for iterates over an iterable.
- range excludes stop.
- A sum starts at 0.
- += reuses the previous value.
- The analytical reference helps validate.

9. Mini-practice

Change one input. Before running, write a comment predicting what should change and why. Then run the file and compare your prediction with the output.

Completion criterion

Move on when you can explain the problem, equation/logic, new Python concepts and execution flow without simply reading the code.